

“abiogenesis”: that life could generate spontaneously from nonliving material. For example, maggots were thought to just appear by themselves in rotting meat. Likewise, infectious diseases were thought to be caused by “miasma” (bad air) rising from decaying matter.

In a famous experiment, Pasteur demonstrated that the air is filled with microbes, and that these will colonize any exposed surface or liquid. First, he filtered air through cotton, examined the cotton through a microscope, and found that it contained the type of microorganisms associated with decaying food. Next, he heated a glass flask of nutrient-rich broth to sterilize it. He then softened the neck of the flask with heat and bent it down and up again into a “swan neck” S-shape. This meant that nothing in the air above the flask could fall directly down onto the liquid in the flask. When the broth cooled, no microbes reappeared in it. However, when Pasteur tilted the

“Everything is clear if its cause be known.”

Louis Pasteur, 1878

neck of the flask so that the outside air could reach the broth, it became contaminated with microbes once more.

Germ theory

After his help with the wine industry, the French government asked Pasteur if he could cure a disease that was destroying silkworms. Pasteur succeeded, having found that it was caused by two different types of parasitic microbe. Now convinced that illness was brought about by certain microbes (germs), he devoted himself to learning more. First, he studied chicken cholera. He cultured

ROBERT KOCH

The German physician Robert Koch was the founder of bacteriology. He discovered the life cycle of anthrax and identified the bacteria that cause cholera and tuberculosis.



After graduating from medical school with flying colors, Koch (1843–1910) soon developed a passion for laboratory work on microorganisms. He devised techniques for culturing bacteria and discovered that the anthrax bacterium produces a type of spore, which can remain dormant in soil. He is particularly recognized for his work on isolating specific microbes: his four “Koch’s postulates” established the key principles for identifying the microbe that causes a disease. In 1905, Koch won the Nobel Prize in Physiology or Medicine for his research into tuberculosis. Once a colleague of Louis Pasteur’s, the two men later became archrivals.

Pasteurization was originally used on wine, but is more familiar to us now as a treatment for milk. Brief heating kills certain bacteria, and the pasteurized milk then stays fresh for 2–3 weeks.

